Evidence-Aware Stagewise Spatio-Temporal Modeling for Auditable Online Interview Proctoring

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Abstract—Suspicious behavior in an online interview is rarely represented by one decisive frame. It is expressed through noisy, temporally distributed evidence such as repeated off-screen gaze, sustained head rotation, an additional face, or a prohibited object. This paper presents an evidence-aware stagewise spatio-temporal framework that converts these cues into reviewable interview segments rather than automatic accusations. MediaPipe Face Mesh and geometric estimators produce head-pose, gaze, eye-openness, face-count, motion, and object-related features. A one-layer attention-based LSTM models 60-frame windows at 15 FPS after per-video z-score normalization. To prevent identity leakage, all recordings from a participant are assigned exclusively to training, validation, or testing. On 794 windows from unseen subjects, the final global model obtains 75.44% accuracy, 73.25% macro-F1, and 65.61% suspicious-class F1. Subject-split learned-stagewise models obtain F1-scores of 55.26% for looking away and 62.72% for offscreen gaze. Their probabilities provide interpretable evidence and, in a decision-support integration experiment, raise macro-F1 to 78.33% and suspicious-class F1 to 73.63%. Face presence obtains 98.58% F1 on 30,000 frames, and multiple-face detection obtains 90.51% F1 on a focused 940-frame test. A separately executed CITW phone-detection experiment reports 98.59% image-level F1. Error and condition analyses show that gaze remains sensitive to camera quality, illumination, glasses, and pose. The framework is therefore intended for human audit and prioritization, not autonomous misconduct adjudication.

Index Terms—Automated proctoring, online interview, spatio-temporal modeling, gaze estimation, LSTM, human-in-the-loop, auditable AI.

I. Introduction

Online interviews are routinely used for recruitment, scholarship selection, professional certification, and academic assessment. They reduce logistical cost and expand access, but they also restrict an evaluator's view to a compressed webcam stream. Assistance can occur outside that view: a candidate may consult a second display, receive prompts from another person, or use a mobile device while remaining visible to the interviewer. Generative-AI tools increase the range of possible assistance, but visual monitoring alone cannot verify which application produced an answer. Accordingly, this study detects behavioral indications that warrant review; it does not claim to identify generative-AI use directly.

Current automated-proctoring systems commonly detect isolated events such as a missing face, a second person, a turned head, an offscreen gaze, or a phone [1], [2]. These events are informative but individually unreliable. A blink may resemble a downward glance, normal thinking may involve temporary head movement, compression may displace iris landmarks, and a face on a poster may be counted as another person. A rule that maps one frame or one geometric threshold directly to misconduct therefore has a high risk of both false alarms and missed events. What matters is not merely whether a cue occurs, but whether several cues persist, repeat, and co-occur over time.

This motivates two complementary levels of analysis. The global level predicts whether a temporal segment should be reviewed. The stagewise level identifies which visual evidence supports that prediction, including face presence, multiple faces, looking away, offscreen gaze, and suspicious objects. The distinction is operationally important. Global classification answers when a reviewer should inspect a session, whereas stagewise analysis helps explain why. Their metrics also have different units—temporal windows, unique frames, and external images—and must not be averaged into a synthetic "overall accuracy."

The stakes are higher in recruitment than in low-risk image classification. A false positive can imply dishonesty, affect employment opportunity, and create psychological or reputational harm. Research on remote proctoring also reports concerns about privacy, fairness, anxiety, and opaque automated decisions [3]–[5]. A defensible system must therefore expose timestamps, supporting cues, confidence, and uncertainty to a human proctor. Its output should prioritize evidence for review rather than impose a final sanction.

This paper develops and evaluates an evidence-aware stagewise spatio-temporal pipeline for that purpose. Each recording is standardized to 640×480 pixels and 15 FPS. MediaPipe Face Mesh provides facial and iris landmarks; geometric modules derive head pose, relative gaze, eye openness, face count, and motion; and YOLOv8 provides object evidence. Fifteen per-frame features form the input of a compact one-layer LSTM with attention. The selected temporal window contains 60 frames (approximately four seconds), advances by 30 frames, and is normalized relative to each recording. Separate learned-stagewise models use 107 continuous and geometric features to predict looking-away and offscreen-gaze behavior without receiving their old binary rule outputs as target shortcuts.

The experimental design addresses a critical source of optimistic bias in behavioral video research. The 60 recordings come from 20 participants, with normal, third-party-assistance, and digital-assistance scenarios for each participant. Splitting individual videos at random would place the same face, camera, background, and movement style in both training and testing. Instead, this work uses a subject-level split: all videos from a participant belong to exactly one partition. The resulting test therefore measures transfer to unseen participants rather than memorization of identity-specific visual patterns.

The contributions of this paper are fourfold:

- an auditable architecture that separates perception evidence, learned temporal behavior, global review prediction, and human interpretation;
- a leakage-aware subject-split evaluation of a one-layer LSTM W60 model, including annotation reliability and explicit minority-class metrics;
- a learned-stagewise evaluation that distinguishes all-video diagnostic performance from unseen-subject generalization and quantifies the value of stagewise probabilities as global decision-support signals; and
- a failure-oriented analysis covering temporal-window sensitivity, feature-group contribution, illumination, camera sharpness, background complexity, and object-detection transfer.

The final global model reaches 75.44% accuracy and 73.25% macro-F1 on unseen subjects. Stagewise-assisted decision support increases macro-F1 to 78.33%, mainly by recovering suspicious windows that the global model misses. These numbers are not presented as proof of autonomous cheating detection. They show that temporal evidence can improve review prioritization while also exposing the remaining generalization gap in gaze-related behavior.

II. Related Work

A. Automated Proctoring and Interview Integrity

Automated proctoring combines identity verification, activity monitoring, and anomaly detection to support remote assessment. Surveys consistently identify face analysis, gaze tracking, head-pose estimation, person counting, audio monitoring, and object detection as common components [1], [2]. However, the literature also identifies persistent weaknesses: behavior is often defined inconsistently, normal movements may be penalized, benchmark datasets are scarce, and reported accuracy can obscure poor minority-class recall. These limitations are directly relevant to online interviews, where natural conversation produces more gaze and pose variation than a fixed-screen examination.

Interview integrity is also broader than conventional exam misconduct. In an asynchronous or live interview, external assistance may involve another person, prepared notes, search engines, a secondary device, or generated text. Prior work shows that generative AI can influence asynchronous interview responses [6], but a webcam-only system cannot establish the origin of content. The observable target must therefore be framed conservatively: visual behavior may indicate that attention has shifted or assistance may be present, but it does not prove which tool was used or whether misconduct occurred.

B. Visual and Temporal Behavioral Evidence

Face visibility and person count are relatively direct visual tasks, whereas gaze and head behavior are more ambiguous. MediaPipe provides a real-time perception framework and dense facial landmarks suitable for estimating head orientation, eye openness, and iris position [7]. YOLO-based proctoring studies detect phones, people, and visible suspicious actions [8], [9]. Interview behavior analyzers similarly combine gaze, head movement, expression, and other nonverbal cues [10]. These systems demonstrate the usefulness of visual evidence, but a frame-level output is not automatically a behavior-level decision.

Temporal learning addresses this mismatch. LSTM networks retain information through gated recurrent state and are effective for sequential behavioral signals [11]. They have been applied to cheating detection and other educational sequence tasks [12]. CHEESE extends the temporal perspective through multiple-instance learning and localization, reporting frame-level AUC on an online-examination dataset [13]. The present work shares the objective of temporal localization but focuses on webcam interview cues, subject-level separation, and an evidence trace that can be inspected by a reviewer.

The use of a compact LSTM is intentional rather than a claim that recurrent models are universally superior. The available dataset contains 20 participants, and the target system must operate on commodity hardware with short update intervals. A compact recurrent model has fewer parameters than a large attention-only baseline and provides a direct sequential representation over a four-second interval. More complex temporal architectures remain relevant future baselines when a larger multi-device dataset is available; they are not reported here because no controlled Transformer experiment is present in the final study.

C. Fairness, Privacy, and Human Review

Remote proctoring processes sensitive visual data and may influence consequential decisions. Empirical studies report that perceived fairness and privacy depend on what data are captured, who receives them, and how decisions are explained [3]–[5]. False positives are therefore not symmetric with ordinary classification errors. In recruitment, an incorrect suspicious flag can affect reputation and opportunity; in education, it can increase anxiety or initiate an integrity investigation.

This paper adopts a human-in-the-loop interpretation. The global classifier produces a priority signal for a time interval. Stagewise modules attach interpretable evidence such as sustained side pose or offscreen gaze. A human proctor remains responsible for considering conversational context, technical quality, and alternative explanations. This design does not remove bias, but it makes the evidence and failure modes more observable than a single opaque verdict.

D. Positioning Against Representative Systems

Table I positions the proposed system against representative work. Direct metric ranking would be misleading because the domains, datasets, labels, and units differ. CHEESE reports frame-level AUC; object-centric systems report detection accuracy; interview-analysis systems predict behavioral traits; and this work reports both window-level global classification and cue-level diagnostics. The meaningful comparison is therefore methodological: temporal context, subject separation, audit evidence, and the intended role of human review.

Three gaps remain visible. First, many systems evaluate videos without ensuring that identities are isolated across data partitions. Second, high aggregate accuracy can coexist with weak suspicious-class F1. Third, a final alert often lacks a measured connection to intermediate evidence. The proposed evaluation is designed around these gaps: subject-level splitting, class-specific metrics, annotation agreement, stagewise generalization, and explicit error analysis.

III. Proposed Method

A. Problem Formulation

Let a standardized interview recording be a sequence of frames $V = \{I_t\}_{t=1}^{F}$ sampled at 15 FPS. The primary task is binary window classification: given a temporal window ending at time t, estimate whether the interval is normal ($y = 0$) or contains behavior that should be reviewed ($y = 1$). Two auxiliary learned-stagewise tasks estimate looking away and offscreen gaze. Face presence, multiple faces, and suspicious-object evidence remain explicit perception outputs. The system does not infer intent or verify use of a particular application.

The architecture follows four evidence layers: (1) visual acquisition and quality standardization, (2) geometric and detector-based feature extraction, (3) temporal learning at global and stagewise levels, and (4) risk presentation for a human reviewer. Fig. 1 shows the final W60 configuration.

B. Video Standardization and Perception

Recordings are resized to 640×480 and decoded at 15 FPS without frame skipping. The working resolution is intentionally moderate: it makes feature extraction consistent across devices while retaining sufficient eye and face detail for the available recordings. MediaPipe Face Mesh produces facial landmarks and iris locations. Six stable facial points are mapped to a generic 3-D face and passed to a perspective-n-point solver to obtain Euler angles $(yaw_t, pitch_t, roll_t)$. Iris centers are normalized relative to eye contours to obtain two-dimensional gaze coordinates $(g_t^x, g_t^y) \in [-1, 1]^2$. Eye aspect ratio (EAR) summarizes eyelid openness.

Frame-to-frame differences capture motion that static geometry misses. The extractor therefore computes changes in yaw and gaze, fixation and saccade indicators, and face-count evidence. YOLOv8 runs as a parallel object module. Because positive phone events are scarce in the interview videos, its capability is evaluated separately on CITW rather than mixed into the main global metric.

C. Feature Representations

The global model receives the 15-dimensional feature vector

$$\mathbf{x}_t = [yaw_t, pitch_t, roll_t, g_t^x, g_t^y, c_t^L, c_t^G, \\ fix_t, sac_t, n_t^{face}, o_t, EAR_t^L, EAR_t^R, \Delta yaw_t, \Delta g_t], \tag{1}$$

where c_t^L and c_t^G are candidate indicators retained as weak global evidence, o_t is suspicious-object presence, and Δg_t summarizes gaze velocity. These candidates are not final behavior verdicts. They are inputs among multiple continuous cues and are interpreted over time by the global model.

The learned-stagewise models use 92 additional Face Mesh geometry variables, yielding 107 features per frame. The expanded representation includes face-box geometry, selected landmark coordinates, relative iris positions, geometric EAR, landmark displacement, and temporal differences. Crucially, the old binary output corresponding to the target is masked. The looking-away model cannot copy a head-angle rule, and the offscreen-gaze model cannot copy a gaze-distance rule. Their outputs must be learned from continuous geometry and its temporal evolution.

D. Temporal Windowing and Normalization

Features are grouped into overlapping windows

$$\mathbf{X}_i = [\mathbf{x}_i, \ldots, \mathbf{x}_{i+T-1}] \in \mathbb{R}^{T \times d}, \tag{2}$$

with $T = 60$, $d = 15$ for the global model, and $d = 107$ for learned-stagewise models. At 15 FPS, W60 corresponds to approximately four seconds. A stride of 30 frames yields a new prediction every two seconds. The number of windows from V videos is

$$N_W = \sum_{v=1}^{V} \left(\left\lfloor \frac{F_v - T}{S} \right\rfloor + 1 \right), \tag{3}$$

where F_v is the frame count and $S = 30$.

Absolute head and gaze coordinates depend on camera placement, seating position, and individual posture. To reduce this recording-specific offset, feature j in video v is standardized as

$$\tilde{x}_{t,j} = \frac{x_{t,j} - \mu_{v,j}}{\sigma_{v,j} + \epsilon}. \tag{4}$$

TABLE I
Macro-level positioning against representative automated monitoring systems. Values are not directly comparable because datasets and evaluation units differ.

Study	Domain	Primary evidence	Temporal treatment	Audit output	Reported emphasis
CHEESE [13]	Online examination	Head, gaze, body, and background events	Multiple-instance and spatio-temporal localization	Localized events	87.58% frame-level AUC on OEP
Behavioral analyser [10]	Interview/viva	Emotion, gaze, head, and smile	Machine-learning behavior analysis	Behavioral traits	Interview behavior and personality analysis
ProctorNet [14]	Online examination	Gaze, mouth, and suspicious actions	Application-level monitoring	Examination alerts	Suspicious-activity detection
YOLO proctoring [9]	Online examination	Devices, persons, face orientation, and audio	Event/action pipeline	Termination or alert events	Visible anomalies and objects
This work	Online interview	Face, head pose, gaze, eye geometry, motion, and objects	One-layer attention LSTM plus learned-stagewise evidence	Timestamp, cue, confidence, and review priority	78.33% macro-F1 for stagewise-assisted review prediction

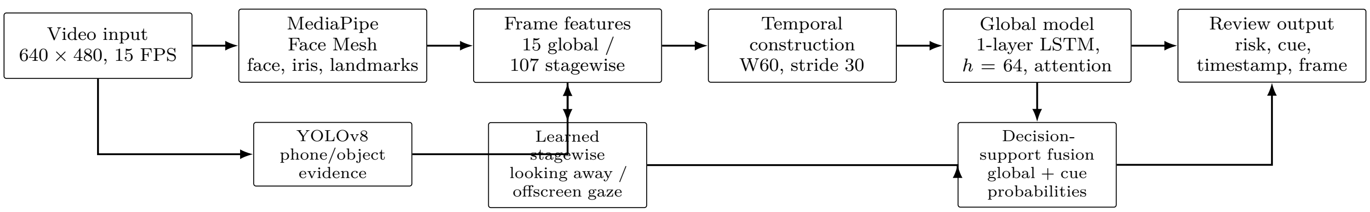

Fig. 1. Final evidence-aware pipeline. The global classifier and learned-stagewise models use the same W60 temporal unit; stagewise probabilities supplement the review decision and provide an evidence trace.

TABLE II
Feature groups and their roles in the final system

Group	Examples	Primary role
Head geometry	Yaw, pitch, roll, face box, landmark coordinates	Looking away
Gaze/iris	Relative iris x/y, gaze velocity	Offscreen gaze
Eye openness	Left/right EAR, geometric eye ratios	Blink and occlusion
Temporal motion	Yaw/gaze change, landmark displacement	Temporal change
Visual context	Face count, eye state, object flag	Auxiliary cue

This normalization uses no class label. It encourages the model to learn deviations relative to a participant's recording rather than a universal absolute pose.

E. Global LSTM with Temporal Attention

The final global network contains one LSTM layer with hidden size 64. For each frame,

$$\mathbf{h}_t, \mathbf{c}_t = \mathrm{LSTM}(\tilde{\mathbf{x}}_t, \mathbf{h}_{t-1}, \mathbf{c}_{t-1}). \tag{5}$$

An additive attention head weights informative positions:

$$a_t = \frac{\exp(\mathbf{v}^\top \tanh(\mathbf{W}\mathbf{h}_t))}{\sum_{k=1}^{T} \exp(\mathbf{v}^\top \tanh(\mathbf{W}\mathbf{h}_k))}, \tag{6}$$

$$\mathbf{z} = \sum_{t=1}^{T} a_t \mathbf{h}_t, \qquad p_G = \sigma(\mathbf{w}^\top \mathbf{z} + b). \tag{7}$$

The classifier uses dropout 0.45 before the final layer. A validation-selected threshold $\tau_G = 0.60$ maps p_G to the suspicious review class. The user interface may display normal, warning, and suspicious bands, but quantitative evaluation is binary so that threshold behavior is unambiguous.

F. Learned-Stagewise Modeling and Evidence Fusion

Looking away and offscreen gaze are trained as separate temporal targets. Each learned temporal model processes the 107-dimensional W60 sequence; the offscreen model additionally uses temporal summary statistics because eye movement is both local and distributional within the window. Their probabilities, p_L and p_O, are not averaged with unrelated frame-level metrics. Instead, they act as aligned W60 evidence that can supplement p_G in a validation-calibrated decision-support layer:

$$p_R = f_\theta(p_G, p_L, p_O, \mathbf{q}), \tag{8}$$

where $\mathbf{q}$ contains face and context evidence available for the same interval. The final output stores p_R, the active cue probabilities, a timestamp, and representative frames. This structure allows a reviewer to trace a high score to sustained gaze, head orientation, face count, or object evidence.

TABLE III
Final subject-level split and W60 sample counts

Partition	Subjects	Videos	W60 windows
Training	14	42	3,719
Validation	3	9	823
Testing	3	9	794

G. Auditable Risk Presentation

The deployment principle is deliberately conservative. The model estimates review priority, not guilt. A risk record contains the temporal score, stagewise evidence, frame-quality indicators, and video location. Short events can be displayed as warnings, while sustained high-probability windows receive higher review priority. A human proctor can then inspect whether the behavior is compatible with normal thinking, a technical failure, or a genuine integrity concern. This separation between measured evidence and adjudication is central to reducing harm from false positives.

IV. Experimental Protocol

A. Participants and Recording Scenarios

The primary dataset contains 60 simulated online-interview videos from 20 participants aged 20–26. Ten participants are male and ten are female; at least five wear glasses. Each participant contributes three recordings: a normal interview, a third-party-assistance scenario, and a digital-assistance scenario. The latter may resemble consultation of generative-AI output, but screen content and answer provenance are not observed. It is therefore labeled as a simulated assistance condition rather than verified generative-AI use.

Recordings include variations in illumination, camera sharpness, distance (approximately 40–80 cm), glasses, partial face occlusion, and background complexity. Original devices support HD-class capture, after which videos are standardized to 640×480 and 15 FPS. The controlled scenarios make event boundaries annotatable, but they do not reproduce the full social and technical complexity of real employment interviews; this limitation is retained in the interpretation of results.

B. Subject-Level Data Separation

All three videos from one participant remain in the same partition. Fourteen subjects (42 videos) are used for training, three (9 videos) for validation, and three (9 videos) for testing. The selected normalized run contains 3,719 training windows, 823 validation windows, and 794 test windows. Validation controls checkpoint selection and the operating threshold; the test partition is used only for final reporting.

This split is stricter than random video division. Identity, room, camera, and personal movement style cannot appear simultaneously in training and testing. Consequently, results are lower than all-video diagnostic evaluation but more representative of deployment to a new candidate.

TABLE IV
Inter-annotator reliability before conflict resolution

Label	N	Agreement (%)	κ
Global state	746	100.00	1.0000
Face present	746	99.33	0.8242
Multiple faces	746	96.65	0.7971
Offscreen gaze	746	90.62	0.7717
Looking away	746	90.62	0.7453
Suspicious object	746	100.00	1.0000

C. Annotation Protocol and Agreement

Two annotators independently label matched two-second base segments. Labels cover the global interview state and five stagewise cues: face presence, multiple faces, offscreen gaze, looking away, and suspicious object. Disagreements are reviewed against written operational definitions before final reconciliation. Agreement before reconciliation is quantified using Cohen's Kappa [15]:

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \tag{9}$$

where p_o is observed agreement and p_e is chance-expected agreement.

Table IV reports 746 matched segments. All labels exceed the predefined 0.70 acceptance criterion. Looking away has the lowest agreement ($\kappa = 0.7453$), consistent with ambiguity between normal thinking and sustained attention shift. Perfect agreement for suspicious objects should not be interpreted as broad object diversity; positive object instances are scarce in the primary recordings.

D. Training Configuration

The final global model is implemented in PyTorch and optimized for binary classification. The one-layer LSTM has hidden size 64 and 24,962 trainable parameters. Training uses dropout 0.45, weight decay 0.002, label smoothing 0.08, Gaussian feature noise with standard deviation 0.025, temporal masking probability 0.45, and feature-dropout probability 0.12. A cosine learning-rate schedule is used. Training ends after 63 epochs; the selected checkpoint is the minimum validation loss (0.6110 at epoch 28). The highest validation macro-F1, 78.03%, occurs at epoch 35, but it is not used as the checkpoint criterion. The final decision threshold of 0.60 is selected only on validation data.

The learned-stagewise models use W60, stride 30, and 107 features to match the global temporal unit. Offscreen gaze additionally uses temporal summary statistics. Both models mask the old target-specific binary signal. Model development also evaluates W15, W30, W45, W60, and W90 to examine the context/responsiveness trade-off.

TABLE V
Final global LSTM configuration

Parameter	Value
Input / temporal shape	60 × 15; stride 30
LSTM depth / hidden size	1 layer / 64
Attention / output	Additive / sigmoid
Dropout / weight decay	0.45 / 0.002
Label smoothing	0.08
Feature noise / feature dropout	0.025 / 0.12
Normalization	Per-video z-score
Checkpoint criterion	Minimum validation loss
Final threshold	0.60, validation-selected

TABLE VI
Evaluation units and valid interpretation

Experiment	N	Interpretation
Global unseen-subject test	794 windows	Final review classification
Stagewise unseen-subject test	794 windows	Cue generalization
Stagewise all-video export	5,249 windows	Diagnostic only
Face-presence test	30,000 frames	General frame-level perception
Multiple-face focused test	940 frames	Positive-case module evaluation
CITW phone presence	1,500 images	External-device object test

E. Evaluation Units and Metrics

The study intentionally uses different evaluation units for different questions (Table VI). The global and learned-stagewise generalization tests use the same 794 W60 windows from unseen subjects. A 5,249-window all-video export is retained only to diagnose cue behavior and enable aligned probabilities across the complete corpus; because it includes training and validation subjects, it is not an unbiased generalization estimate. Face presence is evaluated on 30,000 unique frames from a separate 11-video module test, while multiple-face detection uses a focused 940-frame set containing mapped positive cases. Phone detection is reported from a separately executed CITW experiment with 1,500 images and 2,300 annotated boxes.

For the positive suspicious class, precision, recall, and F1 are

$$P = \frac{TP}{TP+FP}, \quad R = \frac{TP}{TP+FN}, \quad F_1 = \frac{2PR}{P+R}. \quad (10)$$

Accuracy is $(TP+TN)/N$. Macro-F1 averages F1 across normal and suspicious classes and is the main global selection metric because it penalizes a model that succeeds only on the majority class. Results include confusion counts so that every reported percentage can be audited.

V. Results and Analysis

A. Global Model Selection and Convergence

Table VII summarizes global W60 development. The two-layer baseline is retained as historical context, but its 860-window test partition contains a different held-out subject set and is therefore not a controlled head-to-head comparison with the final 794-window run. The optimized and normalized one-layer models share the same final test subjects. Adding per-video normalization increases accuracy from 71.91% to 75.44%, macro-F1 from 68.21% to 73.25%, and suspicious recall from 51.55% to 63.92%. This is consistent with Eq. 4: relative deviations transfer better than absolute pose and gaze coordinates tied to camera placement.

Fig. 2 shows the final 63-epoch training history. Training loss continues to decrease, whereas validation loss reaches its minimum at epoch 28 and then fluctuates. Training macro-F1 also exceeds validation macro-F1 late in training. This gap indicates increasing specialization to training participants, so the minimum-validation-loss checkpoint is preferable to the last epoch. It also supports the decision to report unseen-subject performance rather than training accuracy.

On the final test partition, the model produces $TN = 413$, $FP = 90$, $FN = 105$, and $TP = 186$ (Fig. 3). The 90 false positives mean that approximately one third of global suspicious predictions require rejection by a reviewer. The 105 false negatives show that subtle assistance patterns can still resemble normal interview behavior. For this reason, the model is appropriate for prioritization but not autonomous adjudication.

B. Learned-Stagewise Generalization

Table VIII reports learned-stagewise results on exactly the same 794 unseen-subject windows. Looking away reaches 55.26% F1, while offscreen gaze reaches 62.72% F1. Both models are learned temporal classifiers: their target-specific old binary rule outputs are excluded. Offscreen gaze benefits from temporal summary statistics but remains sensitive to iris localization and eye-region quality. Looking away has higher accuracy but lower positive precision because the normal class is larger and modest head turns are ambiguous.

The all-video diagnostic export gives substantially higher F1: 74.41% for looking away and 85.61% for offscreen gaze over 5,249 W60 windows. Fig. 4 makes the distinction explicit. The all-video values are useful for examining signal coverage and producing aligned probabilities, but they include subjects used during model development. They are not evidence that the model generalizes at that level. The 19.15-point and 22.89-point F1 drops on unseen subjects quantify the actual domain-shift problem.

This gap is an important result rather than a reason to substitute the larger numbers. Gaze and head behavior are

TABLE VII
Global W60 model development. The final two rows use the same 794-window unseen-subject test partition.

Configuration	Layers	Test N	Threshold	Accuracy	Macro-F1	Prec. S.	Recall S.	F1 S.
Historical baseline	2	860	0.50	68.49	61.38	50.93	40.00	44.81
One-layer optimized	1	794	0.53	71.91	68.21	64.63	51.55	57.37
One-layer normalized (final)	1	794	0.60	75.44	73.25	67.39	63.92	65.61

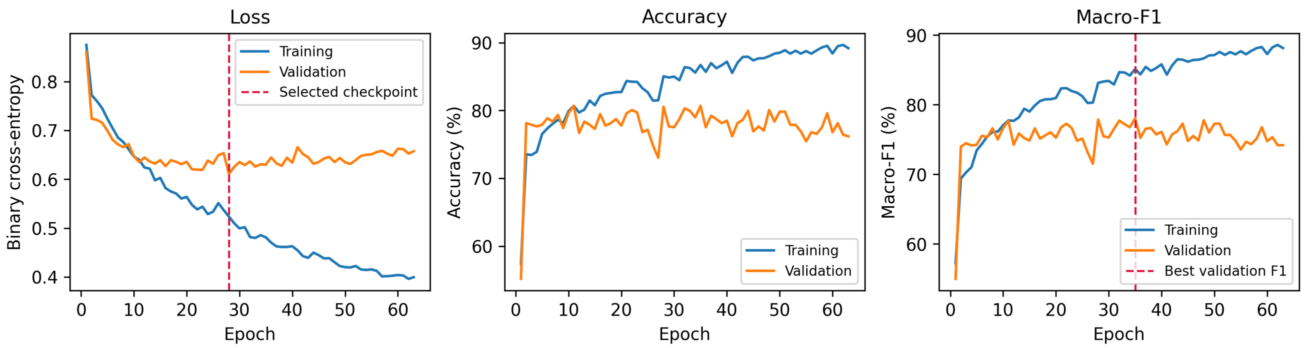

Fig. 2. Training history of the final one-layer normalized LSTM W60. The selected checkpoint minimizes validation loss at epoch 28; peak validation macro-F1 occurs at epoch 35.

TABLE VIII
Learned-stagewise W60 generalization on the final unseen-subject test partition

Target	N	TP	TN	FP	FN	Accuracy	Precision	Recall	F1
Looking away	794	113	498	98	85	76.95	53.55	57.07	55.26
Offscreen gaze	794	164	435	88	107	75.44	65.08	60.52	62.72

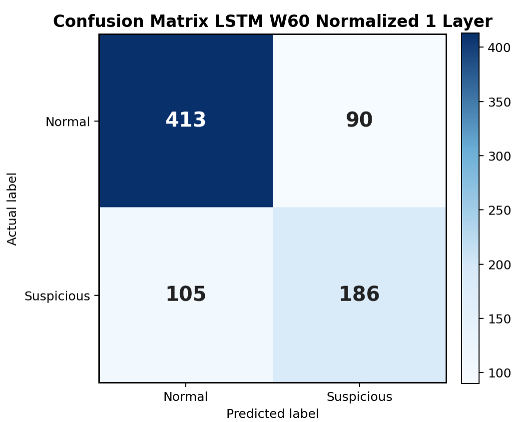

Fig. 3. Confusion matrix of the final global W60 model on 794 windows from unseen subjects.

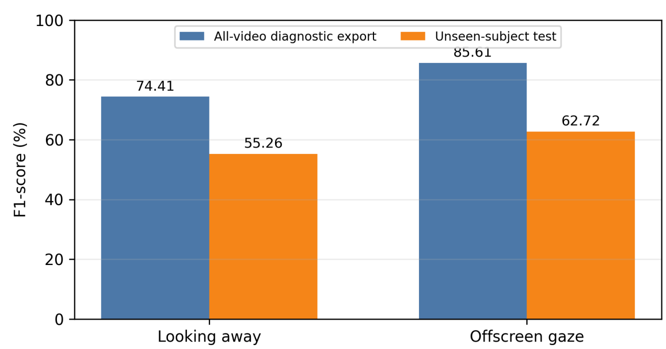

Fig. 4. Diagnostic all-video F1 versus unseen-subject F1. Only the orange bars estimate generalization.

personalized: participants differ in neutral pose, eye shape, use of glasses, camera angle, and natural thinking patterns. Subject-level evaluation exposes that dependence and identifies where additional participants and device diversity are most valuable.

C. Face and Suspicious-Object Modules

Face presence is evaluated on 30,000 unique frames, whereas multiple-face detection uses a focused 940-frame set containing mapped positive cases (Table IX). Face presence reaches 98.58% F1. Multiple faces reaches 90.51% F1 with 43 false positives and 75 false negatives. The result is strong for a perception module but context remains necessary: a poster, photograph, person passing in the background, or a very small secondary face can change the count without implying assistance.

The primary videos contain too few positive object cases for a meaningful end-to-end estimate. A separately executed CITW experiment therefore evaluates phone

TABLE IX
Face-module results under their respective evaluation sets

Module	N	TP/TN	FP/FN	Prec.	Recall	F1
Face present	30,000	29050/114	216/620	99.26	97.91	98.58
Multiple faces	940	563/259	43/75	92.90	88.24	90.51

TABLE X
Separately executed phone-detection evaluation on CITW

Level	N	TP	FP	FN	Prec.	Recall/F1
Image	1,500	1,395	15	25	98.94	98.24/98.59
Object	2,300	2,010	240	290	89.33	87.39/88.35

presence at confidence 0.20 and IoU 0.5. The final thesis reports 98.59% image-level F1 and 88.35% object-level localization F1 (Table X). The difference is expected: localization penalizes duplicate predictions and imperfect box overlap even when phone presence is correctly recognized. Because the complete external prediction dump is not available in this workspace, this experiment is treated as supplementary evidence rather than part of the reproducible global result.

D. Do Stagewise Probabilities Help the Global Decision?

Stagewise metrics are not averaged into the global result. Instead, aligned W60 probabilities are introduced as evidence variables in the decision-support layer of Eq. 8. Table XI compares this configuration with the unchanged global W60 output on the same 794 windows. Accuracy rises by 3.91 points, macro-F1 by 5.08 points, and suspicious F1 by 8.02 points. The largest change is suspicious recall, from 63.92% to 78.69%.

False negatives decrease from 105 to 62, while false positives increase from 90 to 102. This is a meaningful trade-off for review prioritization: more true suspicious intervals are surfaced, but additional normal intervals must be dismissed by a human. The result does not mean that stagewise modules are individually more accurate than the global model. Their value is complementary information about the type and persistence of visual deviation.

E. Temporal-Window Sensitivity

Fig. 7 compares diagnostic learned-stagewise experiments from W15 to W90. Looking-away F1 is 50.57% at W15, falls to 44.32% at W30, and peaks at 59.29% at W60. Offscreen-gaze F1 is lowest at W45 (32.35%) and highest at W90 (51.47%). These experiments have different numbers of windows because longer intervals produce fewer samples, so small differences should not be interpreted as a controlled paired significance test.

W60 is selected as the common operating point for three reasons. First, four seconds is long enough to distinguish a brief glance from a sustained shift. Second, stride 30 still provides an update every two seconds. Third, global

Fig. 5. Representative phone detections from the separately executed CITW evaluation.

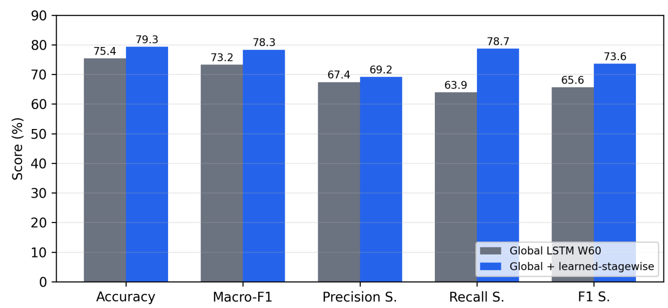

Fig. 6. Effect of adding aligned learned-stagewise probabilities to the global review decision.

and stagewise probabilities become temporally aligned, avoiding an additional resampling policy during fusion. W90 is useful for offscreen gaze but represents six seconds and provides only 652 diagnostic test windows, compared with 988 at W60.

F. Feature-Group Contribution

The final thesis includes a descriptive contribution analysis over extracted feature groups. It is not presented as a retraining-based ablation: no causal claim is made from removing one group. Relative indices in Table XII show that face geometry and landmark stability are strongest for both targets. Gaze and iris features rank second. Head pose contributes more to looking away than to offscreen gaze, while eye openness provides context for distinguishing gaze shifts from blinks.

The low contribution of supporting binary indicators to offscreen gaze reinforces the no-shortcut design. The learned model should rely on continuous gaze, iris, geometry, and temporal dynamics rather than reproduce a threshold-coded candidate. A future causal ablation should retrain every feature subset under the same subject split and equal evaluation anchors; the current descriptive index is useful for interpretation but does not replace that experiment.

VI. Discussion

A. Visual-Condition Analysis

Table XIII compares learned-stagewise F1 after grouping W60 windows by illumination and background complexity. The dim subset unexpectedly outperforms the

TABLE XI
Global decision performance before and after learned-stagewise evidence integration

Configuration	N	Accuracy	Macro-F1	Prec. S.	Recall S.	F1 S.	TN	FP	FN	TP
Global LSTM W60	794	75.44	73.25	67.39	63.92	65.61	413	90	105	186
Global + learned-stagewise	794	79.35	78.33	69.18	78.69	73.63	401	102	62	229

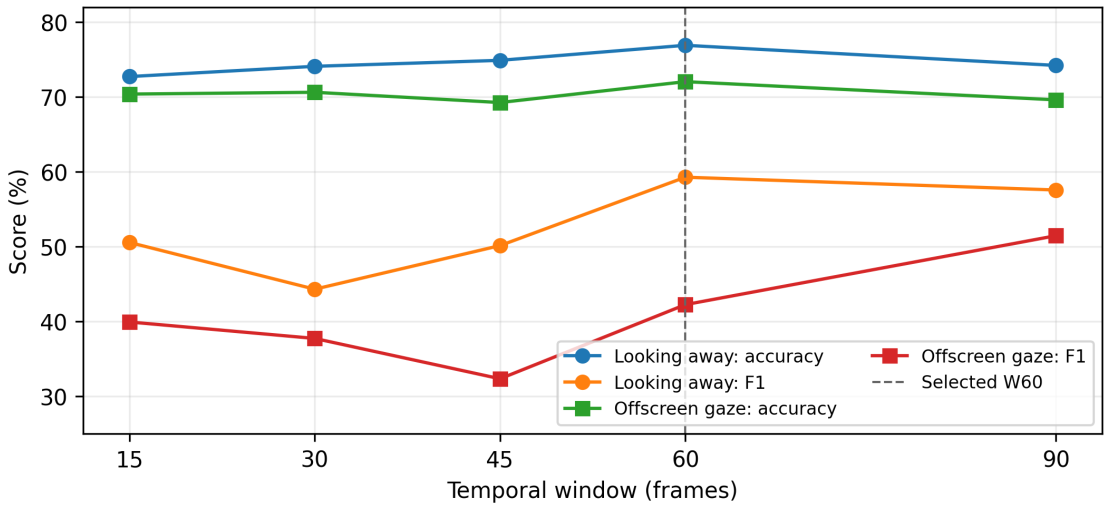

Fig. 7. Diagnostic sensitivity to temporal context. W60 provides the best looking-away F1 and preserves a two-second update interval; W90 improves offscreen-gaze F1 but increases latency and reduces sample count.

TABLE XII
Relative feature-group contribution index (maximum normalized to 100)

Feature group	Looking	Offscreen
Face geometry and landmarks	100.0	100.0
Gaze direction and iris	85.8	90.6
Head pose	80.2	75.3
Eye openness	57.2	55.3
Fixation and saccade	28.7	44.4
Supporting visual indicators	49.0	12.3

bright subset. This does not establish that dim light is beneficial: the groups differ in participant, behavior, and class distribution. The result instead demonstrates why visual-condition analysis must report confounding. Frame inspection still shows that low illumination and blur reduce the stability of iris and EAR measurements even when classification happens to be easier in that subset.

Camera sharpness is assessed qualitatively using Laplacian variance. Sharp recordings preserve iris boundaries and eyelid contours; blurred recordings increase noise in gaze, EAR, and frame differences even when the main face remains detectable. Busy backgrounds have approximately an order of magnitude higher edge density than plain examples and can introduce face-like structures. Thus, face presence remains robust, gaze is dominated by eye-region quality, and multiple-face interpretation is most sensitive to background semantics.

B. Qualitative Evidence and Error Taxonomy

Fig. 8 presents two representative positive sequences. The offscreen-gaze example shows a mostly frontal head with a sustained eye-direction shift. The looking-away example shows a clear side profile. Temporal presentation is important: an isolated crop does not reveal persistence, whereas adjacent frames allow a reviewer to distinguish a blink or transient glance from a sustained pattern.

The principal error modes are summarized in Table XIV. Several are not solvable by threshold adjustment alone. For example, stronger gaze sensitivity may recover subtle offscreen events but also label blinks and normal reading-like eye movement. Likewise, a lower face threshold may recover small secondary faces but detect posters. Temporal models reduce isolated errors, but better calibration data and explicit quality features are still required.

C. Fairness, Auditability, and Deployment Boundary

The system's strongest result is not a justification for automatic sanctions. Even the stagewise-assisted configuration produces 102 false-positive windows. Consecutive windows overlap by 30 frames, so several alerts may describe the same behavioral episode; a reviewer interface should merge them into one event, display uncertainty, and retain the underlying clip. Candidates should be informed about collection, retention, and review policy, and an adverse decision should remain appealable.

TABLE XIII
Observed stagewise performance across visual-condition groups. These are observational subsets, not controlled interventions.

Factor	Condition	N	Looking F1	Offscreen F1	Technical observation
Illumination	Bright	3,077	70.72	79.25	Stable landmarks; heterogeneous behavior
Illumination	Dim	767	84.27	94.72	Lower EAR stability; easier subset distribution
Background	Plain	1,313	77.82	90.33	Fewer face-like distractors
Background	Busy	1,313	71.60	93.42	Lower looking-away recall; more face-count risk

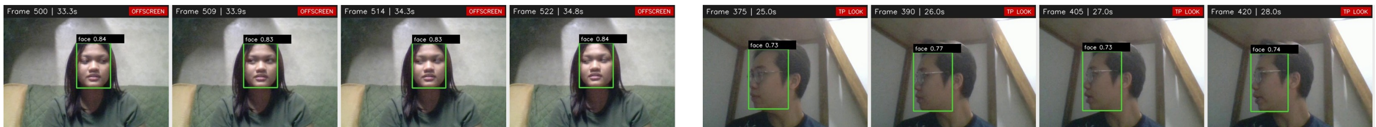
(a) Sustained offscreen-gaze evidence. (b) Sustained looking-away evidence.

Fig. 8. Representative temporal evidence shown to a reviewer. Bounding boxes confirm face tracking; adjacent frames expose persistence.

TABLE XIV
Observed error modes and mitigation directions

Module	Failure mode	Mitigation
Offscreen gaze	Blink, downward thought, iris noise	Quality-aware calibration; more devices
Looking away	Natural head motion, side pose ambiguity	Personalized neutral pose; longer context
Multiple faces	Poster/photo or tiny background face	Temporal tracking and liveness evidence
Phone detector	Small, occluded, or duplicate boxes	Video-domain fine-tuning and tracking
Global model	Subtle assistance resembles normal behavior	Additional modalities and human review

Auditability also requires preserving evaluation boundaries. Frame-level face F1, window-level global macro-F1, and image-level phone F1 answer different questions. Reporting their arithmetic mean would create a number with no valid statistical unit. The global 794-window result is therefore the end-to-end review metric; cue-level results explain component behavior; CITW measures external detector capability. This separation follows the practical requirement that evidence must remain traceable to the task on which it was measured.

D. Threats to Validity and Open Problems

Dataset realism: scenarios are simulated and the participant population is limited. Behavior may change when real employment consequences are present. The digital-assistance scenario cannot be equated with verified generative-AI use.

Generalization: subject-level separation removes direct identity leakage but does not establish cross-institution, cross-device, or demographic fairness. The stagewise generalization gap shows that more participants and camera types are needed.

Evaluation dependence: overlapping windows are not statistically independent. Future work should report episode-level metrics, participant-level bootstrap intervals, and external-video testing. The window-sensitivity experiments also use different sample counts and should be repeated with equal temporal anchors.

Modal coverage: the current system is visual. It cannot determine whether a pause reflects cognition or external prompting, and it cannot observe screen content. Audio, speech timing, screen provenance, and device telemetry could improve evidence, but each introduces additional privacy obligations.

Model comparison: the compact LSTM is appropriate for the current data scale, but a controlled comparison against temporal convolution and a compact Transformer remains necessary. Such models should use the identical subject split, preprocessing, window anchors, and threshold-selection policy; otherwise apparent gains may reflect protocol differences rather than architecture.

VII. Conclusion

This paper presented an evidence-aware stagewise spatio-temporal framework for online interview proctoring. A one-layer attention LSTM operating on normalized W60 sequences obtains 75.44% accuracy, 73.25% macro-F1, and 65.61% suspicious F1 on participants excluded from training. Learned-stagewise looking-away and offscreen-gaze models reach 55.26% and 62.72% F1 under the same strict split. Although these cue models are imperfect, their aligned probabilities increase global macro-F1 to 78.33% and suspicious F1 to 73.63% in the decision-support integration experiment.

The main conclusion is methodological. Subject separation, cue-level reliability, confusion counts, visual-condition analysis, and explicit human review are necessary for credible proctoring research. The system can prioritize evidence-bearing segments, but it cannot establish intent or prove use of generative AI. Deployment should therefore remain human supervised, transparent, and limited until broader external validation is available.

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